# KWAMENKRUMAHUNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

## COLLEGE OF ENGINEERING

DEPARTMENT OF MECHANICAL ENGINEERING

# [MME 291: ECONOMICS AND COMMERCIAL GEOGRAPHY]

### PART A: LECTURE NOTES

**D. SAKYI**

**OUTLINE**

- Unit 1: Basic Concepts in Economics
- Unit 2: Demand and Supply Analysis

Unit 1

# BASIC CONCEPTS IN ECONOMICS

## Introduction

This unit provides an introduction to the study of economics and how economic principles and models aid economic agents in their decision-making. Economics is broadly divided into two: microeconomics and macroeconomics. Microeconomics is concerned with individual parts of the economy while macroeconomics deals with the economy as a whole. We are constantly being made aware of national and international economic problems. Talk of fuel price increases, unemployment, striking of medical doctors, interest rate and exchange rate changes, and economic recession among others. All these are micro-and macro-economic issues, many of which we cover in this course.



## Learning Objectives

After reading this unit you should be able to:

a. appreciate why we study economics
b. distinguish between the concept of scarcity, choice and opportunity cost
c. understand the production possibilities curve and how it can be used to explain opportunity cost and the law of increasing cost
d. distinguish between economic efficiency and equity as microeconomic objectives
e. distinguish between the various economic units and economic systems

# Unit content

1.1 Economic Decision Making

1.2 Scarcity, Choice and Opportunity Cost

1.2.1 The Production Possibility Curve

1.3 Microeconomic and the Economic Problem

1.3.1 Positive Economics versus Normative Economics

1.3.2 Economic Goods versus Free Goods

1.3.3 Economic units

1.3.4 Economic Systems

**1.1 Economic Decision-Making**

The study of economics as a subject can be traced back to the publication of a book entitled *"An inquiry into the Nature and Causes of the Wealth of Nations"* by Adam Smith in the year 1776. This explains why Adam Smith is often referred to as the *"father of economics"*. Yet, a single and precise definition of the subject to cover all its aspects has been difficult. Various economists have given different definitions according to their own thoughts and perceptions of the subject. The earlier definitions of economics often focused on wealth creation, in order to improve the standard of living of mankind through increased consumption. A good example is the definition given by A.C. Pigou. He defined economics as "the means of studying how total production could be increased so that the standard of living of people might be improved".

Modern definitions of economics however, emphasize the fundamental problem of *scarcity* and *choice*. Among these, is the definition given by Professor Lionel Robbins. He defined economics as "the science that studies human behaviour as a relationship between *ends* and *scarce means* which have *alternative uses*". Several important issues are embodied in this definition. Economics studies the behaviour of mankind (i.e. it's a *social science*). It is concerned with *ends* which include all the needs, wants, targets or aspirations of mankind. It is also concerned with scarcity and resources (i.e. our means). That is, economics deals with situations in which resources available are not sufficient to meet all our needs. Finally, the definition recognizes the fact that resources have alternative uses for people to choose from (i.e. *choice*).

Following these definitions we can say that economics is about making *choices* under conditions of *scarcity*. Choices have to be made because the resources needed to produce goods and services are limited in supply, compared to the competing uses for them. The common element in economic

decision making therefore is that *choice involves cost*. Thus, in choosing anything, you must be prepared to sacrifice something else. You could decide to stay in your house and watch TV all day instead of your decision to attend lectures, and preparing to write mid-semester and end of semester examinations for economics. When making these choices, economists usually assume that individuals try to maximize the benefit they receive from the sacrifices they make.

Economics can therefore be defined as the study that ascertains the most efficient allocation of scarce resources among competing uses to ensure increasing production such that living standards of mankind improves through increasing consumption. This implies that the study of economics concerns maximizing society's satisfaction through the production and consumption of goods and services. Therefore, questions such as how many goods and services individuals, firms and the economy as a whole produces, how many people are employed in producing these goods and services, how much do people buy and consume of these goods and services, and how people's consumption are affected by prices among others, are of great concern to economists.

**1.2 Scarcity, Choice and Opportunity Cost**

Everything that is considered valuable is scarce – human skills, goods, time, and money – whereas human wants are *unlimited*. Human behaviour is such that no one is ever satisfied with what he/she has. Everyone at any point in time will like to have more of one thing or the other. Thus, human wants are *unlimited*. Scarcity means that human wants are infinite but the resources available to satisfy those wants are limited. If you had the opportunity to ask both poor and rich people whether they would want more money in the coming month, virtually all of them would say they would want more money so that they can purchase more goods and services. Though our wants are unlimited, our means of fulfilling these wants are limited. This is so because the world has only a limited amount of resources or factors of production (i.e. land, capital, raw materials etc). Moreover, a poor man who cannot afford basic needs like food, clothing, shelter etc will see scarcity as more of a problem than a rich man who cannot afford to build a second house or purchase a second Toyota Land Cruiser. For this reason, individuals, firms and the society at large do not necessarily face the problem of scarcity to the same degree. The key issue however, is that people both rich and poor, want more than they can actually have at any point in time. Economics study the behaviour of mankind in this way.

Because resources are scarce choice has to be made between their alternative uses. In any society, decisions have to be made regarding *what* goods and services are going to be produced and in what quantities, *how* are the goods and services going to be produced - what production resources to use and in what quantities, what production techniques to use -, *for whom* are the goods and services to be produced – how will the output (national income) produced be distributed or what will be the wages and salaries of workers. These microeconomic choices or decisions (be it *rational* or not) has to be made by all societies be it individuals, firms or government. *Rational choice* or *decision* means weighing up of the *costs* and *benefits* of any activity, whether it be firms choosing what and how much to produce, workers choosing whether to work extra hours, or consumers

choosing what to buy. In economics, it is rational to do any activity if benefits exceed the costs of undertaking such an activity. Making these choices or decisions involves sacrifice. Therefore, the production or consumption of one good or service involves the sacrifice of alternatives. This sacrifice of alternatives in the production or consumption of a good or service is what economists call *opportunity cost*. That is the opportunity cost of using any good or service is the value of the next best alternative forgone. Thus, when individuals decide how to use scarce resources, they do so on the basis of opportunity cost. Every scarce good has an opportunity cost, because the production of every good or service uses resources that could have been used to produce something else. Thus, the opportunity cost of working overtime, for example, is the leisure you have to sacrifice. The opportunity cost of buying an economics textbook is the new pair of shoes that you will have to forgo.

### 1.2.1 The Production Possibilities Curve

We have stated earlier that the decisions on how we use our scarce resources are influenced by the concept of opportunity cost. Consider the opportunity cost faced by a KNUST MSc. Industrial Finance and Investment student Ben, who is also an employee of a bank. The items Ben includes in his expenditure plans during weekends that he has to attend lecture are travel costs, meals, entertainment, and a host of other expenses. Although Ben has made a very good budget, he has left out an important expense as he has not included all the costs involved in going for lectures on weekends, while others included are not relevant. By going for lectures on weekends, Ben is giving up what he could earn on working at the bank on weekend (i.e. on Saturdays, which usually pays more). This is the *opportunity cost* which ought to be added to his expenses. Also, he has included expenditure for entertainment which he would incur anyway whether he chooses to go for lectures on weekends or not, especially if this is something he usually does e.g. going to the cinema in the night.

One way we can represent the choices that individuals and society as a whole face is to use a production possibilities curve. A *Production Possibility Curve (PPC)* describes the various combinations of final goods or services that could be produced in a given time period with all available resources and technology fully and efficiently employed. Thus, the PPC shows the various possibilities for allocating scarce resources. It illustrates the concepts of scarcity, choice and opportunity cost. Scarcity is implied by the unattainable combinations beyond the curve; Choice by the need to choose among the attainable points on the curve; Opportunity cost by the downward slope of the PPC.

Figure 1.1 illustrates the behaviour of PPC if an economy can produce only two goods; food and clothing. We measure the output of food and the output of clothing on the vertical and horizontal axes respectively. The curve shows all the combinations of the two goods if they can be produced by employing efficiently the country's resource. It shows a trade-off between the production of food and clothing. At one extreme, the country will be at point A, if it devotes all its resources to

produce food. This will imply it will not produce any clothing. On the other extreme, if it decides to devote all its resources to produce clothing it will be at point D, and will not produce any food. Production cannot take place beyond the PPC (e.g. point F) because the country does not have enough resources to produce at F. The country can produce at point E, but this will represent inefficient use and/or underemployment of available resources. Point E also implies that the economy is producing less of both goods than it is possible for it to produce. We observe that if the country decides to produce more food (say moving along the PPC from point D to C), it would have to sacrifice the production of some clothing. This sacrifice of clothing which enables the production of more food is the *opportunity cost*. Thus, the opportunity cost to the country for producing extra food is the amount of clothing that it has to forgo.

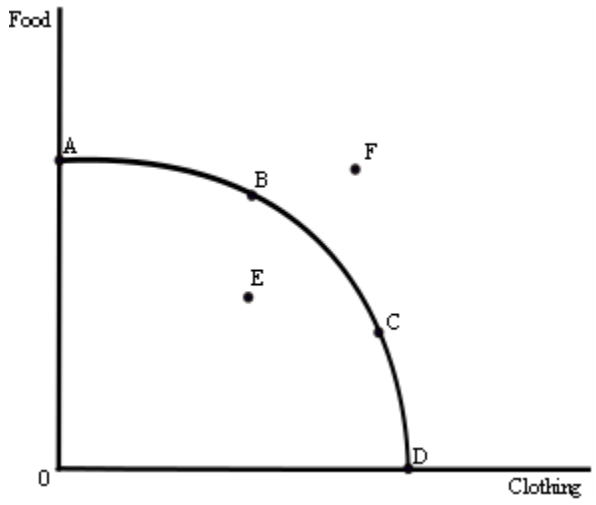



**Figure 1.1: Production Possibilities Curve**

The PPC is bowed outward because the opportunity cost of producing more and more units of food increases as we move upwards along the curve. That is each additional unit of food costs the country more forgone alternatives (clothing) than the previously produced unit. We can see this in Figure 1.2. Although each increment of food is the same, the sacrifices made (i.e. the amount of clothing forgone) in producing these increases (as seen by the arrows). This implies that additional production of one good involves increasing sacrifices of the other or the opportunity cost for each additional unit produced increases at an increasing rate. This is *the law of increasing opportunity cost*. The reason for this is that some resources are better suited for the production of some goods than others as different factors of production have different properties. For example, skills differ among people, land differs in different parts of the country, and raw materials differ from one another. Thus, as countries produce more and more of one good; it has to start using resources that are less and less suitable. It is because opportunity costs increase that the PPC is bowed outward.

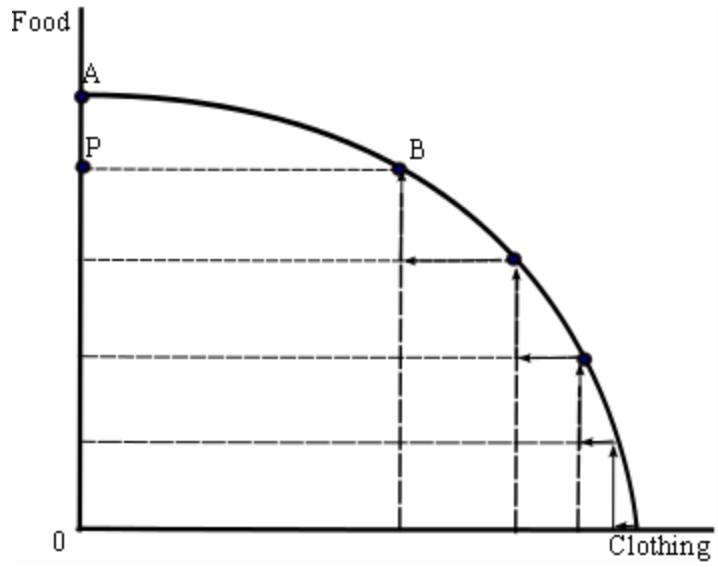



**Figure 1.2: The Law of Increasing Opportunity Cost**

We have noted that if an economy is at the same point on an unchanging PPC, having more of one good necessarily implies having less of the other good. It is however, possible to have more of everything for two reasons (i) when resources previously unemployed are now employed, resources previously used inefficiently are now being used efficiently. This will result in moving from E to B or to C on the PPC in Figure 1.1; and (ii) when economic growth shifts the production possibility curve (PPC) outward. For example, over time investment in new plant will increase the capital stock; new and more efficient raw materials may become available, technological progress are more likely to take place, and through education and training labour may become more productive. All these factors will allow the potential output to increase, and is illustrated by outward shift in the PPC in Figure 1.3.

Specialisation and division of labour are also important. Specialization means that workers provide one type of service. Here, one can be a doctor, an engineer or financial accountant for a firm. Division of labour, on the other hand, occurs when there is specialization within a single industry. For example, in the making of cars, there are car body builders, sprayers, engine tuners, etc. The existence of specialization and division of labour in work allows exchange. This is important because specialists are not self-sufficient and they must exchange what they produce in return for other goods and services they need. Money is a medium of exchange which permits these specialists to buy other goods and services. Specialization and division of labour both allows greater production efficiency. By getting a worker to specialize in just one part of the production process, he is able to develop expert skills, and will be able to work more efficiently, produce more output with less waste of resources.

This analysis is important because, in the discussion of the PPC, we have assumed that resources are fixed over the time period under consideration. Thus, if the PPC moves outward to the right, it

means that the economy is capable of producing more goods and services in total than it could before and there is economic growth. Figure 1.3 illustrates an outward shift of the PPC which allows more of both goods to be produced.

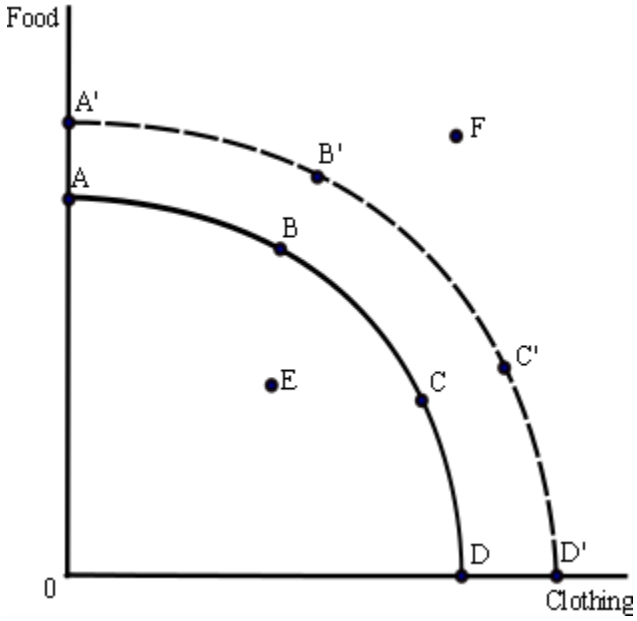



**Figure 1.3: An Outward Shift in the Production Possibility Curve**

### 1.3 Microeconomics and the Economic Problem

The word *micro* is derived from Greek origin which means *'small'*. Microeconomics is concerned with very specific questions or decisions involving individual markets, firms, industries and consumers. How do petrol taxes affect petrol sales? How much will the price of yam increase if floods destroy half the yam crop? How many students will choose not to go to university or polytechnic if the students' loan scheme is abolished? How do consumers decide on what goods to buy and how do firms decide on what goods to produce and what prices to charge are all crucial microeconomic decisions? Microeconomics also analyse market structures; why some markets are highly competitive, with hundreds or even thousands of firms producing a product whereas other markets are very concentrated with only one or only few firms producing a given product. Microeconomics analyses the cause and effects of these market structures as well as positive economic analysis of governments; how and why governments behave the way they do. The study of microeconomics helps us to understand all important issues.

The study of microeconomics helps us to answer the *what*, *how* and *for whom* questions. But how satisfactorily these questions will be answered will depend on the society's objectives. These objectives are broadly grouped into *economic efficiency* and *equity*. *Economic efficiency* is achieved when each good is produced at the minimum cost and where individuals and firms get the maximum benefits from their resources. This implies that economic efficiency would occur when what is produced could make us all better off (or at least some better off without making

anyone worse off). For the society as a whole, what will be considered *economic efficiency* must involve efficiency in production (i.e. *productive efficiency*), and efficiency in consumption and specialization and exchange (i.e. *allocative efficiency*). *Equity*, on the other hand, is where the resulting income out of production is distributed in a way that is considered to be just or fair. Here economists are interested in whether the allocation system is just or fair. Thus, although production and consumption efficiency may be achieved, they might be considered as unfair if some people are rich while others remain poor. In this case there will be no equity. This is clearly a normative issue since the decision made depends upon a value judgment. This concept of equity though difficult to analyze enable us to distinguish between horizontal and vertical equity. Horizontal equity involves equal treatment of equal need. This implies that if the allocation system is to be equitable then it must treat two people with similar problems exactly the same. Vertical equity, on the other hand, involves how we treat unequal individuals differently. For this reason, more often we pay attention to people with more acute medical problems than those without.

### 1.3.1 Positive Economics versus Normative Economics

In economic analysis, distinction is option made between *positive* (*what is*) and *normative* (*what ought to be*) statements. Economics uses positive analysis or statements, such as if *A*, then *B*. Thus, economists when conducting a positive analysis describe the economy and study what causes what. Positive economics is free of value judgment about whether outcomes are good or bad and it involves the study of what does happen. Positive statements are statement of facts which may be true or false but its accuracy can be verified by appealing to the facts. For example, if the price of a good or service goes up relative to the prices of all other goods or services, then the demand for that good or service will fall. When the government raises taxes on fuel, it also includes predictions of what happens to fuel prices and sales, the profits of fuel companies and the amount of tax revenue collected by the government.

Normative economics, on the other hand, is the study of what ought to be where basically people's values come into the analysis. It focuses on whether policy outcome is desirable or undesirable and what, how or whether to change things to achieve the *'best'* possible outcome. Normative economics depends centrally on value judgement concerning what outcomes are *'best'* and it involves the study of what should happen. Normative economics considers the possible benefit that raising cigarette tax might bring better public health, lower heath care cost, more productive work force, etc. Normative economics is usually employed in the formulation of economic policy to tackle issues of positive economics. Hence, it is also called *policy economics*.

### 1.3.2 Economic Goods versus Free Goods

Economic goods (or services) are those which are scarce in relation to the demand for them. That is goods produced from scarce resources. This definition encompasses almost all the resources in the world, except free goods (or services). Free goods are those produced from goods that are not necessarily scarce. The only good which fits somehow well into this category is clean air (mostly

from mountain areas). There is no price or scarcity involved in their use. Given that economist are only interested in goods that are scarce they are generally not interested in free goods. It is important to note that, even goods with natural abundance could be made scarce if some were able to establish property rights over it and hence restrict their supply.

### 1.3.3 Economic units

To make a systematic study of individuals and decision takers more manageable, we consolidate them into three important groups, called economic units. They include *Household*, *Business Firms* and the *Government*.

*Household* is broadly defined as all the people who live under one roof and who take joint financial decisions. Economic theory assumes that: each household takes consistent decisions as if it were composed of a single individual, each household seeks to maximize what is variously called its satisfaction, or its wellbeing or utility, households are the principal owners of factors of production. They sell the services of factors of production and receive their incomes in return. That is, *Households* receive income (wages and salaries) and make several expenditures on consumer goods.

A *firm* is defined as the unit that employs factors of production to produce commodities that it sells to other firms, to households or to the government. Economic theory assumes that; each firm takes consistent decisions as if it were composed of a single firm, each firm takes its decisions with respect to a single goal - to make as much profit as it possibly can, firms are the principal users of the services of factors of production. In factor markets, firms do the buying and household do the selling. Thus, *Business firms* get their revenues from the sale they make to households, other business firms and government. They in turn incur expenditures on the use of factor resources in the form of payment of/for wages, rents, raw materials etc.

*Government* includes all the public agencies, government bodies and other organizations belonging to the central and local government. *Government* obtain its revenue through taxes collected and other fee charges. It then makes expenditures on the provision of social services including those on health, education, road, security services, and other social welfare support systems.

### 1.3.4 Economic Systems

We have noted earlier that all economies are faced with the problem of scarcity and choice. Government, firms and households are also faced with how to answer the fundamental questions of resource allocation (i.e. what to produce, how to produce and for whom to produce). However, the ways these issues are tackled by government differ from one country to another. This is important as government control differ depending on the economic system under consideration.

At the one extreme we have a *free market economy* (a *laissez faire* economy). Here, government normally does not play an active role in the economy. All decisions and choice about resource

allocation are left to market forces of supply and demand, and the workings of the price mechanism. This implies that what producers will make and what consumers will buy are kept in balance by the price that producers will want from their output and the price that consumers are willing to pay for what they purchase. This is achieved through the working of the *price mechanism*. In this economy, ownership and control of factors of production are in the hands of individuals and private firms (i.e. the private sector). The main role of government is to facilitate the operations of the system by maintaining an effective security and legal system with basic social infrastructure in place for the smooth running of businesses.

In a free market economy, there is no need for costly and complex bureaucracies to coordinate economic decisions because of the automatic workings of the price mechanism. Competition among buyers and sellers leads to efficient use of national resources and helps to minimize the central economic problem of scarcity. There is freedom of choice regarding how incomes are spent. However, the free market economy is also plagued with problems. The dominance of few giant firms in the industry often limits the level of competition and the efficient and effective working of the price mechanism. This may lead to unequal distribution of income and under- or non-production of public goods and over production of demerit goods.

At the other extreme is the *centrally planned economy*. Here all the economic decisions and choices about resource allocation are taken by the government. Although, money values are attached to resources and the production of goods and services, it is the government that decides what resources should be used, how much should be paid for them, what goods should be produced and what their prices should be. Thus, government plans the output of each industry and firm, the technique that will be used, and the labour resources required by each industry and firm as well as the distribution of the output between consumers. Ownership and control of resources are also vested in the state in trust for the people.

With the central planning, high growth rates could be achieved if large amounts of resources are directed into investment; consequently, unemployment is avoided and the national income is more equally distributed. However, these goals can only be achieved at enormous social and economic costs, as it is costly collecting and analysing information needed for planning; cumbersome bureaucracy; there is inefficient use of resources due to price controls by government; and among others, there is a considerable loss of individual choice due to state control over resource allocation.

The fact that free markets can fail and government has to intervene to rectify the various failings of the free market gave rise to a third economic system, which is a hybrid of these two extreme economic systems. This economic system is called the *mixed economy*. Here, decisions and choices are made partly by free market forces of demand and supply and partly by government. It is important to note that all economies are mixed, although with different degree of free market and centrally planned decision-making. The government role here may be to control the relative prices of goods and inputs, through taxation or the provision of subsidies; control of relative incomes by the use of taxes or welfare payments; and perhaps the control of the pattern of

production and consumption through the provision of public goods. The fact that government intervenes in this economy can serve to rectify the failures of free market, though its actions (because governments are not perfect) may even bring adverse effect as well.



## REVIEW QUESTIONS

1. Why is the concept of scarcity so important to the definition and study of economics and microeconomics?
2. Give one definition of economics and explain its importance
3. What is the opportunity cost to you of taking this economics course?
4. Define opportunity cost and under what circumstance will opportunity cost be zero?
5. Distinguish between normative and positive economics and give two instances each
6. List the various economic systems and give three characteristics of each
7. What is production possibility curve (PPC)? With the help of a diagram illustrate a PPC of a hypothetical nation showing areas where resources are fully utilized, underutilized and where it is unattainable for the nation to achieve

Unit 2

# DEMAND AND SUPPLY ANALYSIS

## Introduction

In this unit, we examine how the forces of demand and supply serve to establish "normal" prices in markets. We show how the equilibrium of demand and supply in the market maximizes the total benefits received by buyers and sellers, as well as, how changes in the demand and supply affects quantity demanded (supplied) and prices of goods and services.



## Learning Objectives

After reading this unit you should be able to:

a. understand the meaning of demand and supply and their determinants;
b. understand the concept of markets, market equilibrium and efficient allocation of resources (in terms of consumer and producer surplus);
c. distinguish between a change in demand and a change in quantity demanded;
d. analyse shifts in demand and supply and how these affect equilibrium price and quantity

# Unit content

2.1 Markets

2.2 Analysing the behaviour of Buyers (Demand)

2.2.1 Individual and Market Demand

2.2.2 Non-Price Determinants of Demand

2.3 Analysing the behaviour of Sellers (Supply)

2.3.1 Individual and Market Supply

2.3.2 Non-Price Determinants of Supply

2.4 Market Equilibrium

2.41 A mathematical Approach to Equilibrium Analysis

2.4.2 Market Equilibrium and Efficient Allocation of Resources

2.4.3 Movement to a New Equilibrium- Shifts in Demand and Supply

2.4.4 Movement to a New Equilibrium- Simultaneous Shifts in both Demand and Supply

**2.1 Markets**

The idea of a market for many people is one of back-to back stalls in a city centre selling anything from fruit and vegetables to clothing materials. However, for economists, the term *market* has a much broader meaning. Economists usually use the term to describe any process of exchange between buyers and sellers. Thus, formerly, a *market* can be defined as any set of arrangements that allows buyers and sellers to communicate and arrange exchange of goods and services. For this reason, there exists market for the buying and selling of houses (the housing market), shares (the stock market), commodities (commodity market) etc. Where no government intervention occurs in the exchange of these goods and services, then we can have a *free market*. Here no one issues commands or provides directions, but instead the market acts in the following ways: (i) they allocate resources to the production of different goods and services (ii) they determine what resource mix to employ, and (iii) they distribute the goods and services among households. It is the *invisible hand* (according to Adam Smith) of self-interest that acts as the guiding force in this market.

Markets reduce the cost of bringing buyers and sellers together and the cost of finding out what is available for sale, its price, its quantity and how it compares to other items for sale. Thus we say that markets reduce the transaction costs of exchange – the cost of time and information required

for exchange. For example, suppose you want to find a job after graduation. One approach would be to go from employer to employer inquiring about job openings. But this would be time consuming and would require much transportation. Alternatively, you could pick up a copy of job advertisement from recruitment centres or from a newspaper. These advertisements reduce the cost of time and information required to bring buyers and sellers together. This is made possible because of Adam Smith's invisible hand.

**2.2 Analysing the Behaviour of Buyers (Demand)**

What determines how much goods people are willing to buy at any particular time? The most important factor is the price of the good. Therefore, the more expensive it is for one to buy the good, the less he will be prepared to buy and vice versa *ceteris paribus*. This behaviour represent what we call the *law of demand* which states that at higher prices, a lower quantity will be demanded and at lower prices a higher quantity will be demanded, *ceteris paribus* (i.e. everything else remaining constant).

This is the case because when the price of the good rises two things happen - relative prices and real income of potential buyers change. Buyers take both of these into account when they react to the price increase. The change in the relative price means that the good is now more expensive when compared with other similar goods and they react to this by choosing to purchase less of it. This condition of price change is referred to as *substitution effect* of a price change. The increase in the price of the good also reduces buyers real income, implying that they can now only buy less than before with their money or nominal income. The way in which buyers react to a change in their real income depends to a large extent on the nature of the good in question. For *normal goods,* their demand increases when income increases and falls when income decreases but price remains constant. However for *inferior goods,* their demand falls when income increases and increases when income decreases. Therefore, a fall in real income further reduces or increases the amount of goods buyers wish to purchase depending on the nature of the good (whether *normal* or *inferior*). These conditions are referred to as the *income effects* of a price change.

The *substitution* and *income effect* of a price change becomes necessary because, by emphasizing the *ceteris paribus* assumption when we talk about demand, we are assuming that other things apart from price are held constant. This is important because price is not the only variable that affects the demand for a good. Several other factors including consumer's income, prices of related goods, taste and preferences, changes in the expectations of future relative prices and the number of potential buyers (or population) in the market are all crucial determinants of demand. For example, if the consumer's income changes while the price of a good is change, then we would not know whether the change in the quantity demanded is due to a change in the price or to a change in income. For this reason, we hold income constant, as well as any other factor that might affect the quantity of the product demanded

This relationship between price and quantity demanded of a good or service is shown graphically (i.e. the *demand curve*) in Figure 2.1. Given this relationship we can formally defined *demand* as

the quantity of a good or service that buyers or consumers are able and willing to buy in a market at a given price during a specified period of time, *ceteris paribus*. *Demand*, therefore, tells us that the quantity demanded of any commodity is inversely related to that commodity's price, *ceteris paribus*, this implying that the price and quantity demanded move in opposite directions. Price goes up, quantity demanded goes down: price goes down, quantity demanded goes up.

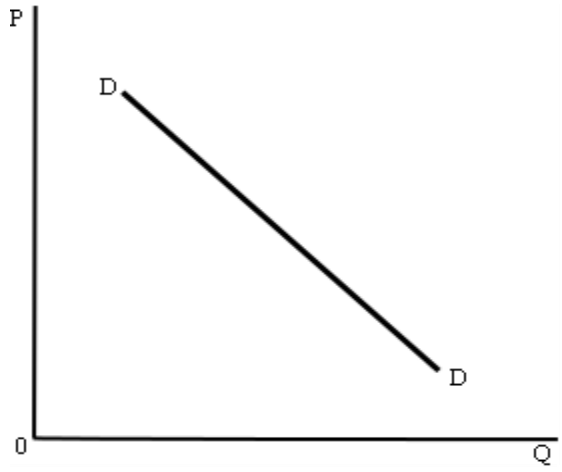
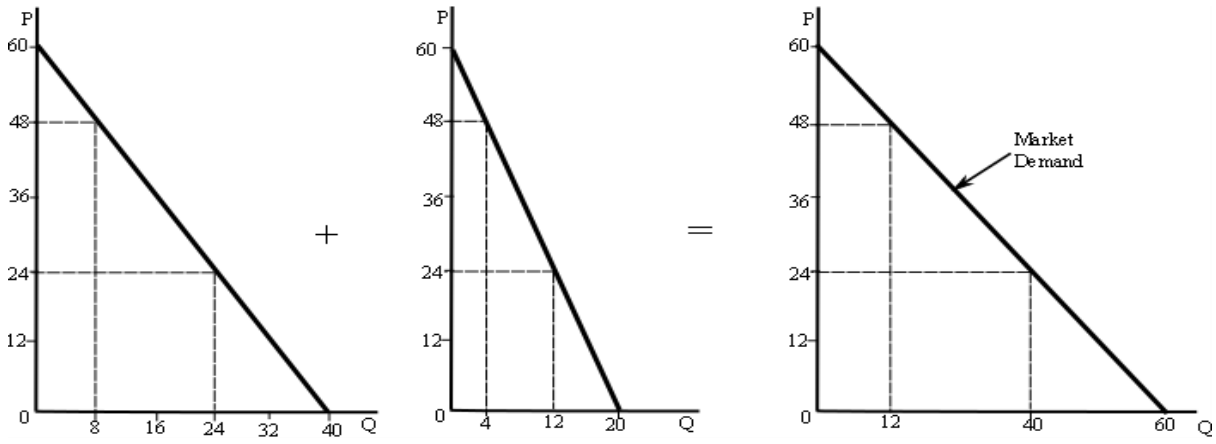



**Figure 2.1: The demand for a good (demand curve)**

### 2.2.1 Individual and Market Demand

It is important that we distinguish between *individual* and *market demand* of a good. The *market demand* of a good or service is simply the sum of the demands of all buyers in a market at a given price during a specified period of time, *ceteris paribus*. Thus, the market demand traces the relationship between the price of a good and the quantity that all buyers in the market are willing and able to buy. This is summarized in Figure 2.2. The demand curve in panel (c) is obtained by summing up horizontally the individual demand curves in panel (a) and (b).




Panel (a) Panel (b) Panel (c)

**Figure 2.2: The market demand for a good**

### 2.2.2 Non-Price Determinants of Demand

The demand curve is constructed on the assumption that "other things remain equal", this implying that none of the determinants of demand, other than price, changes. This assumption implies further that the price of a good is not the only factor that determines the demand for the good. Changes in a number of factors can also cause demand for a good to either increase or decrease *independently* of price. These factors include income, distribution of income, taste, preferences, the number and price of substitute goods, and the number and price of complementary goods, etc

People's demand for most goods, particularly *normal goods* will rise as their income rises. There are exceptions to this rule, particularly for *inferior goods*, because for these goods as people get more income, they tend to spend less on inferior goods, and rather switch to better quality goods. If the national income were redistributed from the poor (rich) to the rich (poor) then the demand for luxuries (inferior) goods would rise. If people find a good more desirable because of its taste then they would demand more of it at each price.

If people's preferences change in favour of (against) a good they will buy more (less) of it at each price. The number and price of substitute (complementary) goods influences the behaviour of the demand curve. The higher the price of substitute goods, the higher will be the demand for this good as people would switch from the substitutes, For example, the demand for Milo will depend on the price of Bournvita. If the price of Bournvita goes up, the demand for Milo will rise, ceteris paribus. Complementary goods are those goods that we normally consume together (e.g. cars and petrol). The higher the price of complementary goods, the fewer will be the demand for the good whose complementary good's price has risen, and vice versa. For example, the demand for polish will depend on the price of shoes. If the price of shoes is to fall, so that more is demanded, then the demand for polish will also rise.

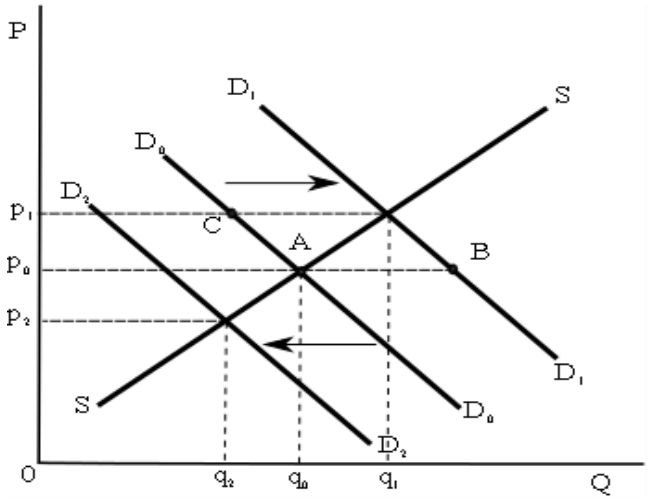



**Figure 2.3: Shifts in market demand for a good**

Whenever there is a change in any of these non-price determinants of demand, the whole demand curve shifts, as shown in Figure 2.3. Assuming$D_0$ is the original demand curve. Then, if a change in one of the non-price determinants causes an increase in demand – say, income rises, for example, the whole demand curve shifts to the right (i.e from $D_0$ to $D_1$. On the other hand, if a change in a determinant other than price causes demand to fall – say, income falls, the whole demand curve shifts to the left (i.e. from $D_0$ to $D_2$). This analysis enables us to understand the distinction between changes in demand (i.e. shifts in the demand curve) and changes in quantity demanded (i.e. movement along the demand curve).

*Changes in demand* result from shifts in the entire demand curve such that the quantity demanded at each and every price changes. The demand schedule or curve is drawn with non-price determinants held constant. However, if it happens that any of the non-price determinants change, then demand curve may shift to the right. This notwithstanding, *changes in the quantity demanded* can result only from a change in price. This implying that changes in the quantity demanded only occurs when there is a movement to a different point along a given demand curve. A change in price causes a movement along a given demand curve.

In sum, *a change in demand* shifts the demand curve and is caused by *changes in the non-price determinants of demand*. On the other hand, *a change in quantity demanded* results from *a movement along a given demand curve* and is caused by a change in price only.

**2.3 Analysing the Behaviour of Sellers (Supply)**

Demand is only one part of how prices are determined in the market for goods and services. The other component is the way in which quantities supplied of goods and services relate to price. If sellers are assumed to want to maximize profits, then they would want to sell more goods at higher prices. For this reason, we can predict a positive relationship between price and quantity supplied.

More formally, *supply* is defined as the quantity of a good or service that sellers are willing and able to sell in a market at a given price during a specified period of time, *ceteris paribus*. Thus, supply establishes a relation between price and quantity supplied and it indicates how much of the good producers are both willing and able to offer for sale at each possible price, *ceteris paribus*. This positive relationship is shown graphically as the supply curve in Figure 2.4.

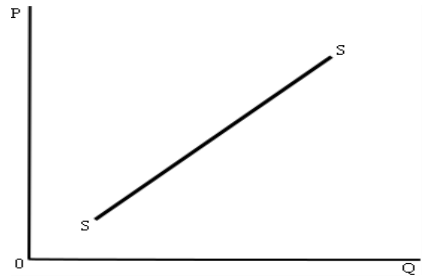
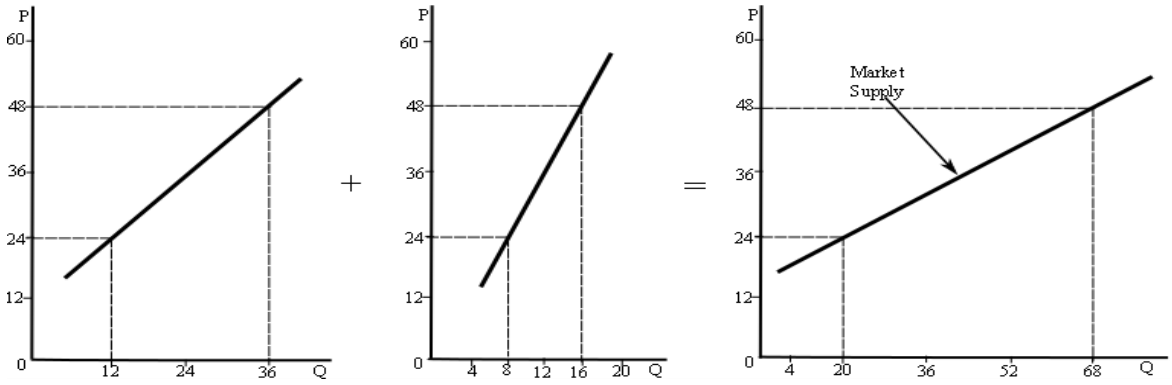



**Figure 2.4:The supply of a good (supply curve)**

The *supply curve* reflects the *law of supply* which states that the quantity offered for sale is positively related to the price; at higher prices, larger amounts are supplied and at lower prices, smaller amounts are supplied, *ceteris paribus*. This is the opposite of the relationship we saw for demand. If price alters there is a *movement along* the supply curve.

### 2.3.1 Individual and Market Supply

Similar to demand, it is important that we distinguish between *individual* and *market supply* of goods and services. *Market supply* of a good or service is simply the horizontal sum of the supply of all sellers in a market at a given price during a specified period of time, *ceteris paribus*. Thus, the market supply traces the relationship between the price of a good or service and the quantity that all sellers in the market are willing and able to sell. This is summarized in Figure 2.5. The supply curve in panel (c) is obtained by summing up horizontally the individual supply curves in panel (a) and (b).




Panel (a) Panel (b) Panel (c)

**Figure 2.5: The market supply of a good**

NB: It is important to note that, not all supply curves are upward sloping. It can be vertical, horizontal or even downward sloping, and this will depend largely on the time period over which firms' response to price changes is considered.

*Why is the supply curve upward-slopping or the positive relationship?*

Sellers offer more goods for sale when prices are higher for three important reasons. Firstly, the higher the price of a good or service, the more profitable it becomes to produce and supply. Thus firms will be encouraged to produce more of it by switching from producing less profitable ones. This profitability provides an incentive for increasing production, and firms become more willing to offer more for sale at a higher price than at a lower price. Secondly, through the theory of increasing costs, firms find that as they supply more they find at a point in time that costs rises more and more rapidly. For example, if a firm supplies manufacturers (i.e manufactured goods), then beyond a certain level of output, costs are likely to rise rapidly as workers have to be paid overtime and machines approach their working capacity. Here producers are only able to increase

the supply of the good at higher prices, because if higher products involve higher costs of producing each unit, producers will need to get a higher price if they are to be persuaded to produce and supply extra products. Lastly, over time, if the price of the good or service remains high, then new producers will be encouraged to set up in producing the same good or service and total market supply will thus rise. It is important to note that whilst the first two affects supply in the short run, the last affect it in the long run.

### 2.3.2 Non-Price Determinants of Supply

Similar to demand, supply is not only determined by price. Several other factors including the costs of production, the profitability of alternative products (substitutes in supply), the profitability of goods in joint supply, the aims of producers, the number of suppliers, and unpredictable events are all important determinants of supply.

The higher the *costs of production*, the less profit firms will make. For this reason, as costs rise, firms will cut back on production and or probably switch to the production of an alternative product whose cost has not gone up that much. The costs of production factor may result from changes in input prices, changes in technology, organisational changes and government policy (which aim at lowering cost). If a substitute product becomes profitable (i.e. *profitability of alternative products*) in supply producers will switch to the production of these substitutes. Here supply of the original good falls. Sometimes when one good is produced, another good is also produced at the same time (i.e. *profitability of goods in joint supply*). These are two goods where the production of more of one leads to the production of more of the other. These goods are said to be jointly supplied. An example is the refining of crude oil which results in the production of both petrol and diesel. If the refinery produces more petrol, due to a rise in demand and hence its price, supply of diesel will as well rise. The supply of some goods depends heavily on nature and random shocks (i.e. *unpredictable events*). For example, weather and disease affecting farm outputs, wars affecting the supply of imported raw materials, the breakdown of machinery, industry disputes, earthquakes, floods and fire etc all can affect the quantum of supply of the good on the domestic market. Lastly depending on *thenumber of suppliers'* available supply will increase if new firms enter the market.

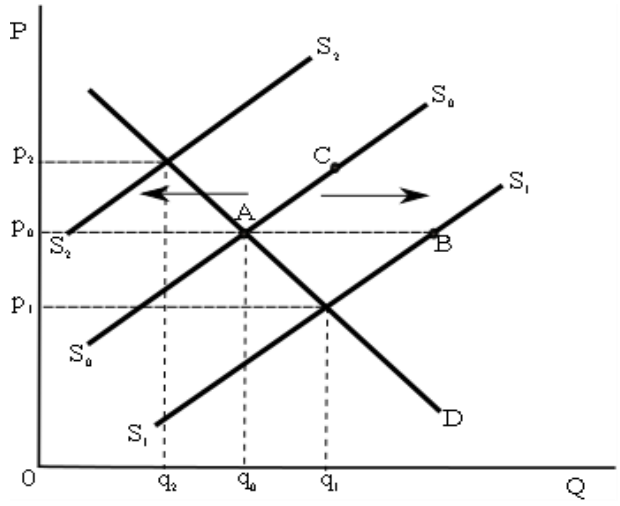


**Figure 2.6: Shifts in market supply for a good**

Whenever there is a change in any of these non-price determinants of supply, the whole supply curve shifts, as shown in Figure 2.6. Assuming $S_0$ is the original supply curve. Then, if a change in one of the non-price determinants causes an increase in supply – say, a decrease in the prices of factors of production (such as land or labour), for example, the supply of the good will increase. This will result in shifts to the right (i.e from $S_0$ to $S_1$) of the whole supply curve. On the other hand, if a change in a determinant other than price causes supply to fall – say, an increase in the prices of factors of production, the whole supply curve shifts to the left (i.e. from $S_0$ to $S_2$). Similarly, if workers' wages increase, supply fall for their service at any given price and the supply curve shifts to the left (i.e. from $S_0$ to $S_2$). If, on the other hand, workers' wages decrease, then the supply curve shifts to the right (i.e. from $S_0$ to $S_1$). Thus, similar to the demand this analysis enables us to understand the distinction between changes in supply and changes in quantity supplied.

A shift in the supply curve is often referred to as a *change in supply*, where as a movement along a supply curve is referred to as a *change in the quantity supplied*. That is, the effect of a price change is illustrated by a movement along the supply curve in Figure 2.6. Therefore, a change in the price of the good itself will cause a movement along the supply curve, implying that if only the price of a good or service changes, its supply curve will not shift. However, if any of the other (or non-price) determinants of supply discussed earlier changes, then the whole supply curve will shift; a rightward shift illustrating an increase in supply and leftward shift illustrating a decrease in supply.

## 2.4 Market Equilibrium

We now analyse the two sides of the market (demand and supply) together in order to determine the equilibrium market price and quantity for goods and services. We show this in Figure 2.7. We observe that there is only one price (i.e.GH₵75) at which the quantity (i.e. 20 units) that buyers want to purchase the good or service equals the quantity that sellers want to sell. This is the *market clearing* or *equilibrium price* and it occurs at the point of intersection of the supply and demand curves. At this equilibrium market clears, leaving no shortage or surplus. This implies that at any other price (aside the equilibrium price) either buyers or sellers are dissatisfied and act to alter the price.

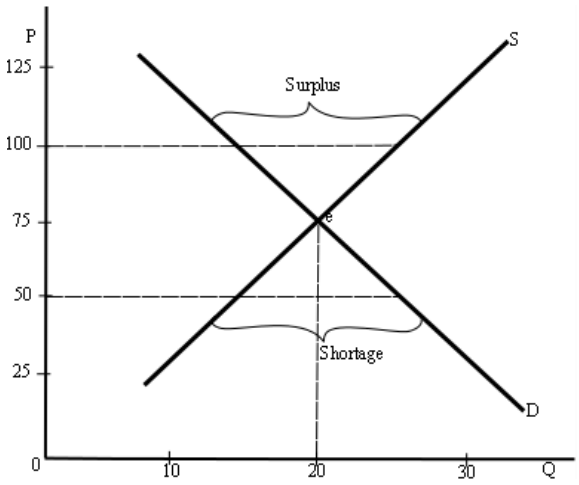



**Figure 2.7: The Determination of Market Equilibrium**

At prices below the equilibrium level, the quantity demanded by buyers is more than the quantity of goods or service that suppliers are willing and able to sell. This results in a temporary shortage and buyers are forced to bid up the price in order to get a few more of the good or service whenever there is some available, until the whole shortage is wiped out. Thus, the actions of both buyers and sellers serve to eliminate the temporary shortage and establish quantity that allows the market to clear.

At prices above the equilibrium level, the quantity of the good or service that sellers wish to supply is greater than the quantity buyers wish to purchase. This results in a temporary surplus. This problem is however eliminated in the market by both buyers and sellers over time as sellers are forced to reduce the price causing them to supply less and buyers to purchase more, until the whole surplus is eliminated. Thus, the actions of both buyers and sellers serve to eliminate the temporary surplus and establish quantity that allows the market to clear.

This analysis implies that both the demand and supply models are very useful for understanding how prices and quantities adjust in the market and how they move towards their equilibrium values at any given point in time.

### 2.4.1 A Mathematical Approach to Equilibrium Analysis

The demand curve (Figure 2.1) and the supply curve (Figure 2.4) in a particular market can be represented by the following two equations:

$$Q_D = a - bP (2.1)$$

$$Q_S = c + dP (2.2)$$

We can determine the market equilibrium price and quantity by equating equations (2.1) and (2.2), since in equilibrium quantity demanded ($Q_D$) equals quantity supplied ($Q_S$).

$c + dP = a - bP$

Subtracting *c* from and adding *bP* to both sides we obtain:

$dP + bP = a - c$

$\therefore (d + b)P = a - c$

$\therefore P = \frac{a-c}{d+b}$(2.3)

We can now solve for the equilibrium quantity by substituting for *P* in either equation (2.1) or equation (2.2).

From equation (1.2)

$$Q_e = a - b\left(\frac{a-c}{d+b}\right)$$

$$= \frac{a(d+b) - b(a-c)}{d+b}$$

$$= \frac{ad + ab - ba + bc}{d+b} = \frac{ad + bc}{d+b} \qquad (2.4)$$

From equation (2.2)

$$Q_e = c + d\left(\frac{a-c}{d+b}\right)$$

$$= \frac{cd + cb + da - dc}{d+b} = \frac{cb + da}{d+b} \qquad (2.5)$$

Therefore:

$$Q_e = \frac{ad + bc}{d+b} = \frac{cb + da}{d+b}$$

### 2.4.2 Market Equilibrium and Efficient Allocation of Resources

Figure 2.8 can help us to understand the idea of *allocative efficiency of market equilibrium* in terms of *consumer surplus* and *producer surplus*. Given that the demand curve measures buyers' willingness to pay ('willingness-to-pay-curve'); it enables us to infer the value they place on what they buy. Any surplus of this value over and above what buyers/consumers actually pay is called *consumer surplus*. Thus, consumer surplus represent the excess of what buyers/consumers are

willing to pay over what they actually pay. This can be computed by finding the area below the demand curve and above the price.

Since the supply curve represents suppliers' minimum supply price, this would represent the benefit sellers get from participating in the market. If the amount received for selling a good is above the minimum amount sellers are willing to accept for the good, there is a surplus to producers. *Producer surplus* thus represent the amount sellers receive for their goods minus the minimum amount they are willing to accept. We can obtain this by finding the area below the price and above the supply curve.

A measure of the sum of *consumer surplus* and *producer surplus* is known as *total surplus* in a market. This measures the benefit to society of a free market outcome. In a free market, the 'correct' quantity of goods is produced in the sense that the total surplus (the sum of consumer and producer surplus) is maximised. This implies that if the quantity is below the equilibrium level the total surplus is not at a maximum and producing more than the equilibrium level of output would lower the total surplus.

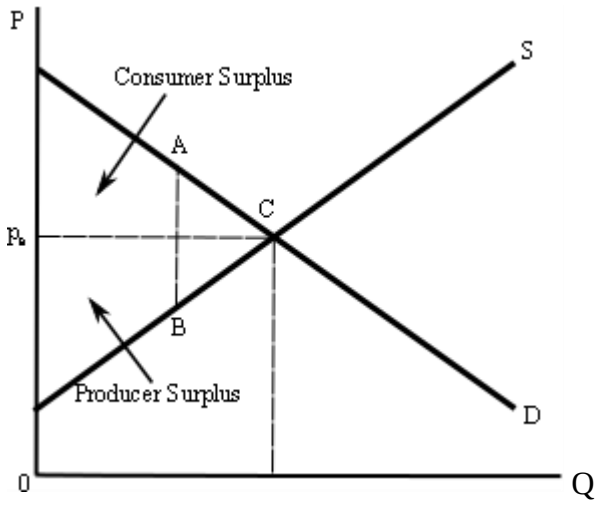



**Figure 2.8: Consumer and Producer Surplus in the Market Equilibrium**

In Figure 2.8, if the quantity is below the equilibrium level of output (say at the level corresponding to line *AB*), the total surplus is not at a maximum, and its size can be increased by the area ABC by raising output to the equilibrium level. On the other hand, the total surplus is lowered when we produce more than the equilibrium level of output. It is only when the equilibrium level of output is produced is total surplus maximised or all resources efficiently allocated.

### 2.4.3 Movement to a New Equilibrium – Shifts in Demand and Supply

The equilibrium price will remain unchanged so long as the demand and supply curves remain the same. Equilibrium is that combination of price and quantity where the desires of demanders and suppliers are exactly matched. Once equilibrium is achieved in the market the equilibrium price

and quantity will remain unchanged unless one of the determinants of demand or supply changes. For example, if either of the curves shifts, a new equilibrium will be formed and the equilibrium price and quantity would change. In Figure 2.9, Panel (a) and (b) illustrates changes in demand while Panel (c) and (d) illustrates changes in supply.

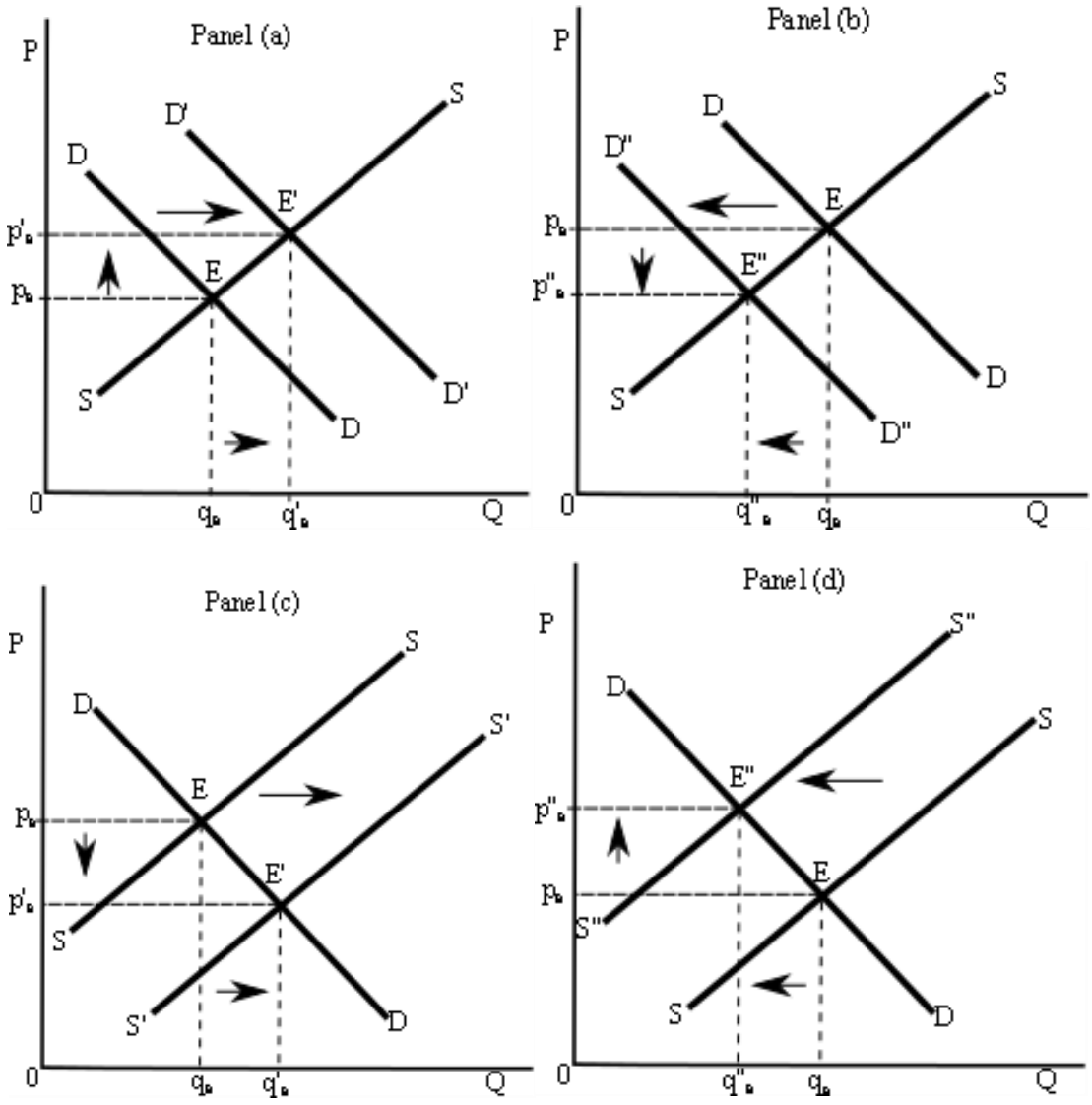



**Figure 2.9: Shifts in Demand and Supply**

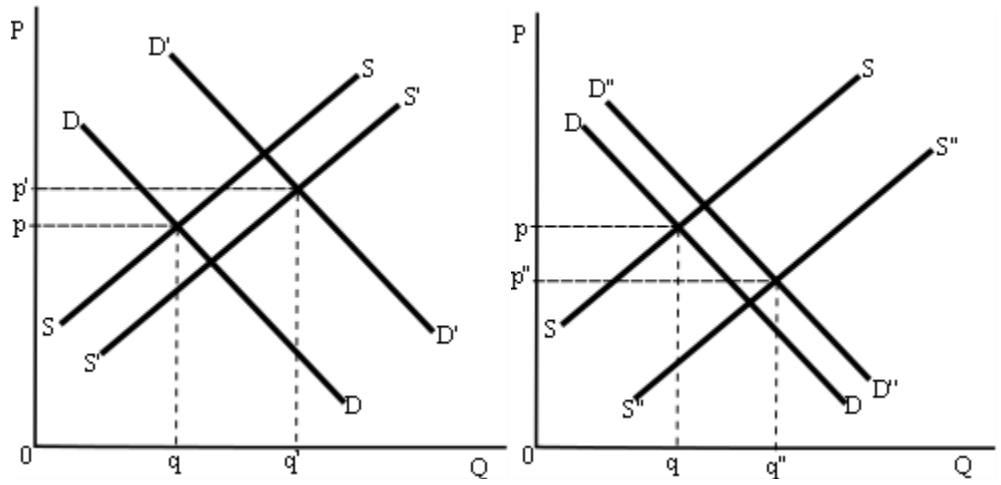
Generally, if the demand for a good or service increases, the equilibrium price and quantity increases as well, given an upward-sloping supply curve. That is if the demand curve shifts and the supply curve remains unchanged, the resulting change in price will move sellers to a different point on their supply curve. The equilibrium price and quantity will move up (see Panel (a)) or down (see Panel (b)) the supply curve as the demand curve either shifts to the right or to the left. Thus, the effect of the shift in demand has been a movement along the supply curve from the old equilibrium *E* to the new equilibrium, *E*' (Panel (a)). At this new equilibrium, which has resulted from the increase in demand, quantity demanded would exceed quantity supplied. The resulting

shortage puts upwards pressure on the price. As the price increase, the quantity supplied increase, but the quantity demanded decrease until once again the two are in equilibrium. This implies that an increase in demand increases both the equilibrium price and quantity. On the contrary, a change in one of the determinants of demands that shifts the demand curve to the left will result in a lower equilibrium price and quantity. As long as the supply curve is upward sloping, any increase or decrease in demand will change equilibrium price and quantity in the same direction as the change in demand. Thus, the effect of the shift in demand has been a movement along the supply curve from the old equilibrium *E* to the new equilibrium, *E*'' (Panel (b)).

For the case of the impact of a change in supply on equilibrium price and quantity, any increase in the supply of the good will lead to a lower equilibrium price and a higher equilibrium quantity, given a downward sloping demand curve. This will come as a result of a shift of the supply curve to the right. On the contrary, a change in any of the factors that result in a reduction in supply will shift the supply curve to the left, causing equilibrium quantity to fall but equilibrium price to rise. This implies that a shift in the supply curve, with the demand curve held constant, will cause equilibrium quantity to change in the same direction as the change in supply but will cause equilibrium price to change in the opposite direction. Thus as the supply curve shifts to the right, price decreases but quantity increases and as the supply curve shifts to the left, price increases but quantity decreases.

### 2.4.4 Movement to a New Equilibrium – Simultaneous Shifts in Both Demand and Supply

Generally, it is easier to predict what will happen to equilibrium price and quantity when only the demand curve or the supply curve shifts at a time as we have seen in the previous section. However, if both curves shift simultaneously, the outcome is not so clear. For example, let's assume that both demand and supply increases; say as result of a decline in the price of Milo which increases the demand for milk. Here we can predict that when this happens, equilibrium quantity increases as well, but what happens to the equilibrium price? We cannot easily determine what will happen to the price unless we consider the effects for specific shifts in the supply and the demand curves. If the shift in demand dominates the shift in supply, then the equilibrium price will increase. On the other hand, if the shift in supply dominates the shifts in demand then the equilibrium price will fall.




**Panel (a)** **Panel (b)**

**Shifts in demand dominates** **Shifts in supply dominates**

**Figure 2.10: Shifts in demand (supply) dominates**

Panels (a) and (b) of Figure 2.10 illustrates this point. In Panel (a) the shift in demand dominates the shift in supply whilst in Panel (b) the shift in supply dominates the shift in demand. We observe that in both panels demand and supply increases in both panels. This results in an increase in the equilibrium quantity for both panels. The equilibrium price however, increases in Panel (a) but decreases in Panel (b). This implies that whenever demand and supply both increase the equilibrium quantity increases, but the effect of an increase in both demand and supply on the equilibrium price depends on which of these shifts (i.e. either demand or supply) dominates the other.

On the contrary, if both curves decrease, the equilibrium quantity will decrease as well, but again, the effect of a decrease in both demand and supply on the equilibrium price will depend on which of these shifts (i.e. either demand or supply) dominates the other. If demand shifts dominate the supply shifts, the price will decrease. The price will however increase if supply shifts dominate the demand shifts.

We can also have a situation where demand and supply move in opposite directions. In this case, we can predict what will happen to equilibrium price: equilibrium price will increase if demand increases and will decrease if demand decreases. However, we cannot easily predict without reference to particular shifts what will happen to equilibrium quantity. For example, if demand increases and supply decreases, the equilibrium price will increase. However, the effect on the equilibrium quantity will depend on which of the shifts (i.e demand or supply) dominates. If demand shifts dominate the supply shifts for an increase in demand which results in an increase in price, then the equilibrium quantity will increase. On the other hand, if the supply shifts dominate the demand shifts for an increase in demand, then the equilibrium quantity will decrease.

## REVIEW QUESTIONS

1. Define the terms; demand, supply, and market equilibrium.
2. Explain four determinants of demand.
3. Distinguish between a change in demand and a change in quantity demand
4. Assume the demand and supply of cassava in Kumasi is given by the table below;

| Price Per Kg (GH¢) | Qty Demanded (thousands of Kg) | Qty Supplied (thousands of Kg) |
|---|---|---|
| 2.00 | 28 | 10 |
| 2.40 | 26 | 12 |
| 3.10 | 22 | 13.5 |
| 3.50 | 19.5 | 19.5 |
| 3.90 | 17 | 22 |
| 4.30 | 14.5 | 23.5 |

   a. Plot the demand and supply schedules on a single set of axis.
   b. Identify the amount of excess demand and supply associated with each price
   c. Identify the equilibrium price and quantity in this market
5. Use a supply and demand diagram to show consumer and producer surplus when a market is in equilibrium. In what sense is this efficient outcome?
6. Consider the demand and supply of rice on KNUST campus to be given by;

$$Q^D = 100 - 3P$$
$$Q^S = 10 + 2P$$

   a. Plot the demand and supply curves.
   b. What is the equilibrium price and quantity of rice at KNUST
   c. Solve for equilibrium price and quantity if the demand function changes to $Q^D = 180 - 3P$